AquaSentinel - 25 New Unique Feature Ideas

Proposed roadmap additions F38-F62 - difficulty-level table with full details | 28 October 2026

1. What this document is

AquaSentinel is an agentic real-time flood response and evacuation optimization system: flood prediction (ML trained on INDOFLOODS), a multi-agent decision layer, QAOA-inspired route choice, ambulance / shelter / boat dispatch, and a digital-twin simulation running on six real OpenStreetMap cities (Delhi, Mumbai, Chennai, Kolkata, Guwahati, Patna).

The existing roadmap NEW_FEATURES.md already holds 37 features: F1-F37 (F35-F37 still planned). The 25 ideas below were chosen so that **none of them duplicates F1-F37**. Every idea obeys the project's rules: real data with a named source, assumed values labelled and swept, no fabricated numbers, one authoritative state, additive artifacts.

Numbering: idea n maps to feature code F(n + 37), so idea 1 is F38 and idea 25 is F62. This is a proposal list - vet an idea before adding it to NEW_FEATURES.md.

2. Difficulty legend

Level	Planning effort	Typical shape of the work
Easy		
Easy-Med	3-5 days	One new service or panel plus wiring and tests.
Medium		
Medium-Hard	8-12 days	New modelling or simulation dimension; needs careful validation.
Hard		

Note: effort values are *planning estimates*, not measured results - the same discipline the project applies to any assumed parameter.

3. Master table - all 25 ideas with difficulty level

#	Feature idea	Category	Difficulty	Data source & assumptions	Builds on (existing code / features)
1	**Drone Recon Fleet**	A. Field ops & fleet	Medium	OSM depots + assumed drone fleet and battery range (labelled)	Boats pattern F22; road closures F1; live map UI
2	**Helicopter / air ambulance**	A. Field ops & fleet	Easy	Assumed fleet, labelled; fires only when no road AND no boat is reachable	Boat pattern F22; hospitals F21; why-card F13
3	**Evacuation buses (mass transport)**	A. Field ops & fleet	Medium	Assumed bus fleet (~40 seats each), labelled; swept in stress test	Shelter allocation F2; dispatch F8; fleet board
4	**Volunteer & NGO coordination board**	A. Field ops & fleet	Medium	Synthetic volunteer registrations, labelled; skills + shifts assumed	Zone warnings F23; queue rows; Agents feed mas.py
5	**Ground truth evidence layer**	A. Field ops & fleet	Medium	Crowd reports (in-repo) fused with satellite mask + model forecast	Satellite check F20; road forecast F10; driver app F3
6	**IoT water-level probes at bridges**	A. Field ops & fleet	Easy-Med	Assumed probes (labelled); coverage swept 0-100 percent	Road flood-time forecast F10; stress test F12
7	**Vulnerable persons pre-registry**	B. Citizens & people	Easy	Synthetic pre-registry, labelled; ticket-only privacy like F31	Zone warnings F23; fairness board F14
8	**Accessible shelter matching**	B. Citizens & people	Easy-Med	OSM wheelchair / toilets tags; fallback tag assumed + labelled	Shelter allocation F2; fairness board F14
9	**Real shelter census (occupancy fix)**	B. Citizens & people	Medium	Stage-machine arrivals + ticket check-ins; fixes the documented gap	SOS tickets F4; shelters F2; replay F5
10	**Citizen push alerts (opt-in)**	B. Citizens & people	Medium	Browser Web Push, opt-in per zone; no external service for local demo	Service worker F15; zone warnings F23; forecast F10
11	**SOS text understanding**	B. Citizens & people	Medium	Synthetic corpus + regex offline; optional LLM behind grounded()	SMS parser F24; AI assistant numeric guard
12	**Operator photo triage wall**	B. Citizens & people	Easy-Med	Already-stored SOS photos + tickets; annotations are operator-only	SOS uploads F4; duplicate merge F30
13	**Demand-capacity gap forecast**	C. Planning & analysis	Medium	Modelled demand from rainfall forecast curve, labelled; capacity real	Rain forecast F18; what-if F16; road forecast F10
14	**Road corridor congestion model**	C. Planning & analysis	Medium-Hard	Assumed capacity per road class, swept; delay grows with stacked units	Routing F1; dispatch F8; QAOA route choice
15	**Dam-breach downstream wave scenario**	C. Planning & analysis	Hard	dams.py physics release pulse; arrival times labelled modelled	Dam registry + reservoir physics; alert thresholds F11
16	**Recovery & return-home planner**	C. Planning & analysis	Medium	Inverse of the closure rule; return trips staged, labelled	Road forecast F10; replay frames F5; shelters F2
17	**Ward damage & impact report**	C. Planning & analysis	Medium-Hard	OSM building footprints + a cited literature depth-damage curve	Building parser from F27; zones + population F19
18	**Post-flood dengue risk map**	C. Planning & analysis	Easy-Med	Published seasonal risk factors; stagnant-water proxy labelled modelled	Zone population F19; OSM water layers
19	**Dam release QUBO optimizer**	D. Intelligence & optimization	Medium-Hard	Synthetic release-schedule cost function; benchmarked like F6 / F12	qaoa_route.py statevector core; benchmark cards

#	Feature idea	Category	Difficulty	Data source & assumptions	Builds on (existing code / features)
20	**Predictive SOS surge model**	D. Intelligence & optimization	Medium-Hard	Poisson rate per zone from rainfall nowcast; validated on replayed runs	Rain forecast F18; what-if F16; replay F5
21	**Dispatcher training mode (serious game)**	D. Intelligence & optimization	Medium	Synthetic blind scenario; skill score vs hindsight optimum, labelled	What-if core F16; hindsight regret F34
22	**Auto SITREP briefing**	D. Intelligence & optimization	Easy-Med	Generated from stored frames; ICS-style template; no invented numbers	Replay store F5; run report.pdf endpoint
23	**Operator copilot with guarded actions**	D. Intelligence & optimization	Medium	Proposes actions; executes only after typed confirmation; each logged	MAS mas.py; ai_assistant grounded(); audit plan F37
24	**Printable zone evacuation pamphlets**	E. Outreach & delivery	Easy	Live city data + QR code to /safe; languages reused from F24	Languages F24; /safe page F15; capacity F27
25	**EOC wallboard / TV dashboard**	E. Outreach & delivery	Easy	Read-only kiosk route; pure frontend over the existing state feed	state() polling feed; all existing panels

Difficulty mix: 4 x Easy | 5 x Easy-Med | 11 x Medium | 4 x Medium-Hard | 1 x Hard
Effort = planning estimate only. Each idea also needs its test pass and a browser check as per the standard feature workflow.

4. Full details - one block per idea

Each block: what it adds, the difficulty, the data story (real source vs labelled assumption) and which existing feature cores it reuses.

A. Field operations & fleet - new rescue capability

1. Drone Recon Fleet [Medium]

When a road is closed or an area is unknown, the operator dispatches a drone from a depot that flies the route and confirms the closure with an aerial photo overlay on the live map. Fleet size and battery range are assumed and labelled, exactly like the F22 boat fleet. Drones give the command centre a sensing agent that reports before any ambulance is committed.

Data: OSM depots + assumed fleet · Builds on: F22 boat pattern, F1 closures, live map UI

2. Helicopter / air ambulance [Easy]

For a critical patient where there is no open road AND no boat reachable, an assumed air-ambulance unit lifts the person directly to a hospital. It reuses the boat dispatch pattern but triggers only in the worst case, and the why-card (F13) explains why air was chosen over road or boat.

Data: assumed fleet, labelled · Builds on: F22 boats, F21 hospitals, F13 why-card

3. Evacuation buses (mass transport) [Medium]

Ambulances move one or two people; buses move about forty. An assumed bus fleet runs shelter-bound runs from warned zones, measured in people moved per vehicle-minute. A new fleet-board column and a big visual payoff during demos.

Data: assumed fleet (labelled) · Builds on: F2 shelter allocation, F8 dispatch, fleet board

4. Volunteer & NGO coordination board [Medium]

Volunteers register with skills - first aid, boat driving, cooking; the agents match them to tasks: door-knocking in warned zones, guiding people at shelters, sandbagging. A shift board with check-in / check-out keeps the current roster honest.

Data: synthetic registrations, labelled · Builds on: F23 warnings, queue rows, mas.py feed

5. Ground truth evidence layer [Medium]

Drivers (F3 app) and citizens report what they see, e.g. water about 40 cm on a given road. Crowd reports fuse with the satellite mask (F20) and the model forecast (F10) into one evidence panel, each source with a confidence tag - the project becomes evidence-aware, not only model-driven.

Data: in-repo reports + existing layers · Builds on: F20, F10, F3

6. IoT water-level probes at bridges [Easy-Med]

Assumed sensor probes at key junctions stream water depth each tick and feed the closure forecast earlier - before water ever reaches the road surface. Sweeping probe coverage from 0 to 100 percent in the F12 stress test shows how much earlier warnings save re-routes.

Data: assumed sensors, labelled · Builds on: F10 forecast, F12 stress test

B. Citizens & people - human-facing

7. Vulnerable persons pre-registry [Easy]

Pre-registered people - elderly, dialysis, wheelchair (synthetic, labelled) - are linked to a zone. When F23 issues a zone warning for that zone, they are auto-priority-flagged in the queue before any SOS arrives. Privacy follows the ticket-only pattern of F31.

Data: synthetic registry + privacy rules · Builds on: F23, F14 fairness board

8. Accessible shelter matching [Easy-Med]

OSM shelters often carry wheelchair and toilets tags. Wheelchair and elderly people are then allocated only to accessible shelters; the fairness board gains an accessibility column; a red alert fires when no accessible shelter is reachable.

Data: OSM tags + assumption fallback · Builds on: F2 allocation, F14 fairness board

9. Real shelter census (occupancy fix) [Medium]

The backend shelter occupancy today is a growth curve unrelated to rescues - CLAUDE.md says so itself. Make occupancy real: rescued people check in with their ticket, occupancy = stage-machine arrivals + check-ins, and the dashboard shows assigned vs arrived vs missing.

Data: stage machine + tickets · Builds on: F4, F2, F5 - fixes a documented gap

10. Citizen push alerts (opt-in) [Medium]

/safe users subscribe per zone in the browser (Web Push through the existing service worker). When their zone gets an F23 warning or a road floods, they hear a real notification instead of pull-refreshing. Proactive alerting with zero external service for the local demo.

Data: opt-in subscriptions · Builds on: F15 sw.js, F23 warnings, F10

11. SOS text understanding [Medium]

Free-text SOS input is turned into structure: three people, one injured, water chest deep becomes people:3, medical:injured, water:chest. Precision and recall are measured on a synthetic corpus the F20 way; a regex path works offline and any LLM path stays behind grounded().

Data: synthetic corpus · Builds on: F24 sms_parse, grounded() guard

12. Operator photo triage wall [Easy-Med]

All SOS photos appear in one wall grid; the operator annotates confirmed, duplicate of X, or needs an area check. No AI claims about photos - verification stays human-in-the-loop and evidence stays with the ticket.

Data: stored uploads + tickets · Builds on: F4 uploads, F30 duplicate merge

C. Planning, scenarios & analysis

13. Demand-capacity gap forecast [Medium]

Predicted SOS arrivals for the next 2-6 hours - modelled from the rainfall forecast curve and labelled - are compared against the real fleet and shelter capacity over the same window. A gap chart turns red hours in advance when demand exceeds capacity.

Data: modelled demand, real capacity · Builds on: F18 forecast, F16 what-if, F10

14. Road corridor congestion model [Medium-Hard]

Road classes get an assumed vehicle capacity; stacking all ambulances on one corridor builds a queue delay that is added to route cost. Routes then carry a waiting-time estimate per road, and the optimizer learns to spread units - a genuinely new dimension for the routing story.

Data: assumed capacity, swept · Builds on: F1 routing, F8 dispatch, QAOA route choice

15. Dam-breach downstream wave scenario [Hard]

Use the existing dams.py reservoir physics to send a release pulse downstream: arrival time at each town, the stage curve, and which city zones flood at which tick. A dramatic, grounded preset - Bhakra releases X, Delhi reaches stage Y four hours later.

Data: existing dam physics · Builds on: dams.py, dam risk service, F11 thresholds

16. Recovery & return-home planner [Medium]

After the peak, the same engine runs in reverse: the inverse of the F10 closure rule shows when each road reopens, zones return in stages, shelters de-occupy with planned return trips. Adds a Phase 2 - Recovery badge, a lifecycle stage the demo currently lacks.

Data: inverse closure rule, labelled · Builds on: F10, F5 frames, F2 shelters

17. Ward damage & impact report [Medium-Hard]

OSM building footprints (already parsed for F27) plus one cited literature depth-damage curve give a per-zone flooded building count and an estimated damage band, downloadable as PDF. Every number is labelled and sourced.

Data: OSM buildings + cited curve · Builds on: F27 parser, F19 zones

18. Post-flood dengue risk map [Easy-Med]

A stagnant-water proxy - flooded low-lying zones with slow drainage - combined with published seasonal dengue risk factors produces a modelled aftermath risk card for health planners. Explicitly marked modelled, not medical advice.

Data: published factors, labelled · Builds on: F19 population, OSM water

D. Intelligence & optimization - the paper-strength area

19. Dam release QUBO optimizer [Medium-Hard]

Schedule spillway releases over 24 hours so the downstream barrage stays below danger level; the cost balances exceedance against energy / irrigation loss. A small QUBO-style scheduling model on the existing QAOA statevector core extends the quantum storyline from routes to a new domain.

Data: synthetic physics, benchmark card · Builds on: qaoa_route.py, F6 / F12 benchmarks

20. Predictive SOS surge model [Medium-Hard]

Rainfall nowcast becomes a predicted SOS rate per zone (Poisson rate model), validated on replayed stored runs. Feeds the what-if planner - surge expected next hour: about 12 calls - and gates the surge button with an estimate instead of a guess.

Data: fit on replay data, labelled · Builds on: F18, F16, F5 replay

21. Dispatcher training mode (serious game) [Medium]

The operator plays a synthetic scenario without seeing the future; at the end the F34 hindsight engine scores their choices against the optimal-in-hindsight, giving a skill score and a where-you-lost-minutes timeline. A teaching feature that no typical final-year project has, and it reuses the F16 + F34 cores almost for free.

Data: reuse of existing simulators · Builds on: F16 ops_sim, F34 regret

22. Auto SITREP briefing [Easy-Med]

One button generates a structured hourly situation report: key numbers, map snapshot, decisions taken, fairness status, regret delta - ICS-style PDF. The per-run report.pdf already exists; a timestamped SITREP chain is a real command-centre practice that is new here.

Data: stored frames · Builds on: F5 replay store, report.pdf

23. Operator copilot with guarded actions [Medium]

A chat-like copilot can propose actions, e.g. close shelter X and add 2 units - each executed only after a typed confirmation and logged line by line, which anticipates the planned F37 audit log. The grounded() numeric guard extends to action proposals.

Data: MAS + AI tab reuse · Builds on: mas.py, ai_assistant, F37 plan

E. Outreach & delivery

24. Printable zone evacuation pamphlets [Easy]

One PDF per zone: shelter list with capacity from F27, a safe-route card, and a QR code straight to the /safe page - generated with reportlab in the F24 languages. Turns the digital system into something physically distributable during a real flood.

Data: live city data · Builds on: F24 languages, F15 /safe, F27 capacity

25. EOC wallboard / TV dashboard [Easy]

A read-only /wall route with giant numbers - rescued, waiting, silent zones, regret score - the live map and the latest agent actions, auto-refreshing for the control-room TV. Pure frontend over the existing state poller; instant demo value at exhibitions.

Data: reuses state feed · Builds on: all existing panels

5. Top 5 if you build only a few

Rank	Idea	Why it is the best pick
1	**13. Demand-capacity gap forecast**	Strongest new analytical idea; reuses F10 / F16 / F18; a great Algorithms-tab card for the IEEE paper.
2	**1. Drone recon fleet**	Most visually impressive demo feature; a clean extension of the F22 boat pattern.
3	**21. Dispatcher training mode**	Genuine uniqueness - very few final-year projects have it; nearly free because F16 + F34 are already built.
4	**9. Real shelter census**	Fixes a documented honesty gap in your own docs; reviewers and examiners value exactly this.
5	**5. Ground truth evidence layer**	Ties F10 (model) + F20 (satellite) + crowd reports into one system-level story.

6. Suggested first build order

Quick wins first (less than a week each): 25 wallboard (F62), 7 pre-registry (F44), 24 pamphlets (F61), 2 air ambulance (F39), 12 photo wall (F49).

Then the reuse-heavy mediums: 13 gap forecast (F50), 22 SITREP (F59), 21 training mode (F58), 16 recovery planner (F53), 9 shelter census (F46), 8 accessible matching (F45).

Then the bigger modelling ideas, one at a time with full tests: 20 surge model (F57), 19 dam QUBO (F56), 14 congestion (F51), 17 damage report (F54), 15 dam wave (F52). Plenty of paper material lives in this last group.

Fit in between as time allows: 1 drones (F38), 3 buses (F40), 4 volunteers (F41), 5 evidence layer (F42), 6 probes (F43), 10 push alerts (F47), 11 text understanding (F48), 18 dengue map (F55), 23 copilot (F60).

Generated as a companion to NEW_FEATURES.md (F1-F37). None of the 25 ideas duplicates an existing or planned feature. Every idea keeps the project's rules: named data sources, labelled assumptions, measured numbers only, additive artifacts, tests after every build.